

Problems Proposed for 2025-26 Pal4Pal

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1 Samuel

1.1 Problem 1

Alice and Bob play a two-player game in which the numbers a and b are initially written on the whiteboard. Starting with Alice, the players take turns subtracting 1 or 2 from one of the current numbers written on the whiteboard. The first player to reach 0 as one of the numbers wins. Given that a and b are chosen randomly between 1 to 101 inclusive, what is the probability that a and b are so that Alice has a winning strategy?

Proposed by Samuel Li

Answer: $\boxed{\frac{6934}{10201}}$

Solution: Denote the ordered pair (x, y) as a winning position if Alice has a winning strategy when $a = x, b = y$ and as a losing position otherwise.

Firstly, $(1, i)$ and $(2, i)$ as well as $(i, 1)$ and $(i, 2)$ are all winning positions for all $1 \leq i \leq 101$, for clear reasons.

Next, notice that if one of $(x - 2, y), (x - 1, y), (x, y - 1), (x, y - 2)$ is a losing position, then (x, y) is a winning position, and if all of $(x - 2, y), (x - 1, y), (x, y - 1), (x, y - 2)$ are winning positions, (x, y) would be losing.

Therefore, we can iteratively find whether (x, y) is winning or losing for all x and y , and it can be trivially proven using induction that (x, y) is losing if and only if $x \equiv y \pmod{3}$.

Therefore, there are $\frac{1}{3} \cdot 99^2$ losing positions (x, y) among $1 \leq x, y \leq 101$. Thus, the probability of choosing a losing position will be $\frac{33 \cdot 99}{101^2}$, so the probability of choosing a winning position will be

$$1 - \frac{33 \cdot 99}{101^2} = \boxed{\frac{6934}{10201}}.$$

1.2 Problem 2

Find the number of primes p for which there exists a positive integer m that is of the form $12k + 2$ such that $48m(47m + 1)$ is p times a perfect square.

Proposed by Samuel Li

Answer: $\boxed{0}$

Solution: Write $48m(47m + 1)$ as $p \cdot x^2$, for some positive integer x . Notice

$$\gcd(48m, 47m + 1) = \gcd(m - 1, 47m + 1) = \gcd(m - 1, 48) | 48.$$

Because $m - 1$ is not divisible by 2 nor 3, either $48m = a^2, 47m + 1 = pb^2$ or $48m = pa^2, 47m + 1 = b^2$, for some a, b with $x = ab$, where $\gcd(a, b) = 1$. Since m is $2 \pmod{4}$, $v_2(48m) = v_2(48) + 1 = 5$, the former is impossible.

In the latter case, consider $47m + 1 = b^2$. Taking mod 12 on both sides,

$$-m + 1 \equiv b^2 \pmod{12}.$$

Therefore, $m \equiv 0, 1, 4, 9 \pmod{12}$, contradicting $m \equiv 2 \pmod{12}$, thus, the answer is $\boxed{0}$.

2 Eric

2.1 Problem 1

Let S be the sum of the squares of the divisors of 480 (including itself). Compute the number of divisors of S .

Answer: $\boxed{108}$

Solution: Since $480 = 2^5 \cdot 3 \cdot 5$, we can see that $S = (1 + 2^2 + 2^4 + 2^6 + 2^8 + 2^{10})(1 + 3^2)(1 + 5^2)$ because if we expand out of the parenthesis, we have every square of each divisor. By geometric series, $S = \frac{4^6-1}{3}(10)(26) = (260)(4^3+1)(4^3-1)/3 = (260)(65)(21) = 2^2 \cdot 3 \cdot 5^2 \cdot 7 \cdot 13^2$. Thus, S has $(3)(2)(3)(2)(3) = \boxed{108}$ factors.

2.2 Problem 2

An ellipse in the first quadrant is tangent to both the x -axis and the y -axis. The coordinates of its foci are $(8, 12)$, and $(v, 12)$. Compute v .

Answer: $\boxed{18}$

Solution: By the definition of an ellipse, the distance between the two foci and the two tangent points on the x and y axes must be the same. Thus, $8 + v = 2\sqrt{(\frac{8-v}{2})^2 + 12^2}$ as the tangent point on the x -axis must be on the perpendicular bisector of the two foci. Squaring both sides we have that $64 + 16v + v^2 = 4(16 - 4v + \frac{v^2}{4} + 144) = v^2 - 16v + 640 \rightarrow 32v = 576, v = \boxed{18}$.

3 Catherine

3.1 Problem 1

Let $\triangle ABC$ be a right triangle with right angle at A and AC the shorter leg. Point D lies inside the triangle such that $\angle ADC = 90^\circ$ and $\angle DCA = \angle ABC$. Extend BD to a point E so that $\angle BEA = 90^\circ$ and $\angle DAE = \angle ABC$. Find the ratio $BD : DE$.

Answer: $\boxed{3}$

Hint: Angle chasing, Angle bisector theorem, Pythagorean theorem.

3.2 Problem 2

Compute $\cos^4 \frac{\pi}{9} + \cos^4 \frac{2\pi}{9} + \cos^4 \frac{3\pi}{9} + \cos^4 \frac{4\pi}{9}$.

Answer: $\boxed{\frac{19}{16}}$

Hint: Note that $\frac{\pi}{9}, \frac{7\pi}{9}$ and $\frac{13\pi}{9}$ satisfy $\cos 3\theta = \frac{1}{2}$. Let $r = \cos \frac{\pi}{9}, s = \cos \frac{7\pi}{9}, t = \cos \frac{13\pi}{9}$, then r, s, t are three roots of the cubic equation derived from $\cos 3\theta = \frac{1}{2}$.

3.3 Problem 3

In triangle ABC , $BC = 6$ and $6 \sin A + 5 \cos B \cos C = 9$. Find the area of ABC .

Answer: $\boxed{6}$

Hint: From $6 \sin A + 5 \cos B \cos C = 9$, one can derive that $B = C$.

Solution: Since $\cos B \cos C = \frac{\cos(B+C) + \cos(B-C)}{2} = \frac{-\cos A + \cos(B-C)}{2}$, the equation

$$6 \sin A + 5 \cos B \cos C = 9$$

becomes

$$6 \sin A - \frac{5}{2} \cos A + \frac{5}{2} \cos(B-C) = 9.$$

Note that $6 \sin A - \frac{5}{2} \cos A \leq \frac{13}{2}, \frac{5}{2} \cos(B-C) \leq \frac{5}{2}$, thus we must have $B = C$ and $\sin A = \frac{12}{13}, \cos A = -\frac{5}{13}$. Then we have $AB = AC = \sqrt{13}$ and can easily get $[\triangle ABC] = \boxed{6}$.